

A SEQUENCE THAT LOOKS BACK HALF WAY

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ABSTRACT. Let $a_1 = 1$, $a_n = a_{n-1}$ for even n and $a_n = a_{n-1} - a_{(n-1)/2}/n$ for odd n . A question on Mathematics Stack Exchange asked for the constant A with $na_n \rightarrow A$, and a commenter conjectured $A = 1/(1 - \ln 2)$ from its digits. We prove this. The proof rests on an exact identity, $na_n = 1 + \sum_{n/2 \leq k < n} a_k$, which turns the recurrence into an average over a window of total weight tending to $\ln 2 < 1$. Every step is formally verified in Lean 4.

1. THE QUESTION

Start with $a_1 = 1$ and build a sequence of rationals by two rules. At an even step nothing happens; at an odd step the sequence drops by a fraction of an earlier term:

$$a_n = \begin{cases} a_{n-1}, & n \text{ even,} \\ a_{n-1} - \frac{1}{n} a_{(n-1)/2}, & n \text{ odd.} \end{cases}$$

The first terms are $1, 1, \frac{2}{3}, \frac{2}{3}, \frac{7}{15}, \frac{7}{15}, \frac{13}{35}$, and the sequence falls like A/n with $A \approx 3.2588913$. The user AAK asked on Mathematics Stack Exchange in August 2023 (question 4748129) for a way to compute A and an expression for it in known constants; the sequence came from a branching process in a random environment. The same day Tian Vlasic searched the digits and suggested $A = 1/(1 - \ln 2) = 3.2588913\dots$, and Gary derived the equation $[(1-x)F(x)]' = 1 - F(x^2)$ for the generating function F and observed that $-\log(1-x)/(1 - \ln 2)$ satisfies it asymptotically as $x \rightarrow 1^-$.

Theorem 1. $\lim_{n \rightarrow \infty} na_n = \frac{1}{1 - \ln 2}$.

This is the coefficient version of Gary's remark, reached without a Tauberian argument. The odd-indexed terms have numerators $1, 2, 7, 13, 281, 530, 29599, \dots$ and denominators $1, 3, 15, 35, 945, 2079, 135135, \dots$; neither sequence is in the OEIS (searched 10 October 2026).

2. THE WINDOW IDENTITY

Let the *window* of n be $W_n = \{k \in \mathbb{Z} : n/2 \leq k < n\}$, so $W_7 = \{4, 5, 6\}$, $W_8 = \{4, 5, 6, 7\}$ and $W_1 = \emptyset$.

Lemma 2. For every $n \geq 1$, $na_n = 1 + \sum_{k \in W_n} a_k$.

Proof. Let $Q_n = na_n - \sum_{k \in W_n} a_k$, so $Q_1 = 1$. Passing from W_{n-1} to W_n adds the index $n-1$, and removes the bottom index $(n-1)/2$ exactly when n is odd. Hence $Q_n - Q_{n-1} = n(a_n - a_{n-1}) + [n \text{ odd}] a_{(n-1)/2}$, which is 0 by the recurrence in both cases. $\square$

For $n = 7$ the lemma reads $7 \cdot \frac{13}{35} = 1 + \frac{2}{3} + \frac{7}{15} + \frac{7}{15}$. By induction it also shows $a_n > 0$: the right-hand side is at least 1 when the earlier terms are positive. Figure 1 shows the identity for $n = 24$.

Write $c_n = na_n$. Since $a_k = c_k/k$, the identity becomes

$$(1) \quad c_n = 1 + \sum_{k \in W_n} \frac{c_k}{k}.$$

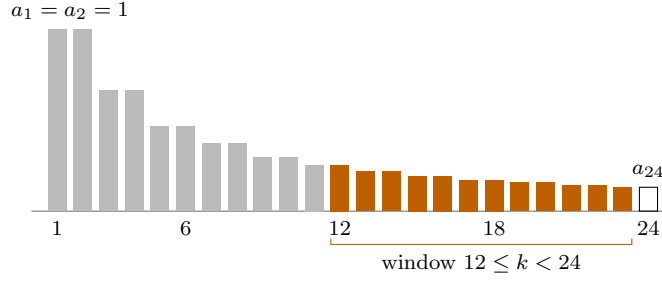


Figure 1. The terms a_k for $k \leq 24$. The copper bars are the window $W_{24} = \{12, \dots, 23\}$. Lemma 2 says that twenty-four copies of the last bar (hatched) exceed the copper bars by exactly 1.

So c_n is 1 plus a weighted sum of earlier values of c , with total weight

$$s_n = \sum_{k \in W_n} \frac{1}{k} \approx \int_{n/2}^n \frac{dx}{x} = \ln 2.$$

If c_n has a limit A , then $A = 1 + A \ln 2$, which is the conjectured constant. The proof makes this honest, and the only thing it needs about the weights is their limit.

Lemma 3. $s_n \rightarrow \ln 2$.

Proof. With $N = n - 1$ and $t = \lceil n/2 \rceil - 1$ the window is $\{t + 1, \dots, N\}$, so $s_n = H_N - H_t$, a difference of harmonic numbers. Since $H_m - \ln m$ tends to Euler's constant,

$$s_n = (H_N - \ln N) - (H_t - \ln t) + \ln \frac{N}{t} \rightarrow \ln 2,$$

because $2t \leq N \leq 2t + 1$. □

3. PROOF OF THE THEOREM

Boundedness. By Lemma 3, $s_n \leq \frac{3}{4}$ for all $n \geq N_0$. Let B be the larger of 4 and $c_1 + \dots + c_{N_0}$. Then $c_n \leq B$ for $n \leq N_0$, and if $c_k \leq B$ for all $k < n$ with $n > N_0$, then (1) gives $c_n \leq 1 + s_n B \leq 1 + \frac{3}{4}B \leq B$. So $0 < c_n \leq B$ for all n .

The error. Let $A = 1/(1 - \ln 2)$, so that $A = 1 + A \ln 2$, and put $e_n = c_n - A$. Substituting $c_k = A + e_k$ into (1),

$$(2) \quad e_n = \sum_{k \in W_n} \frac{e_k}{k} + A(s_n - \ln 2),$$

so if $|e_k| \leq R$ throughout the window, then

$$|e_n| \leq s_n R + \delta_n, \quad \delta_n = A |s_n - \ln 2|.$$

For $n \geq N_0$ this is at most $\frac{3}{4}R + \delta_n$, and $\delta_n \rightarrow 0$. The error in the window shrinks by a factor $\frac{3}{4}$, up to a vanishing term, each time we pass to indices twice as large.

Proof of Theorem 1. The errors are bounded, say $|e_n| \leq M$. Given $\eta > 0$, choose N_1 with $\delta_n \leq \eta$ for $n \geq N_1$. We claim that for each $j \geq 0$ there is an N with

$$|e_n| \leq \left(\frac{3}{4}\right)^j M + 4\eta \quad (n \geq N).$$

For $j = 0$ this holds with $N = 0$. If it holds for j with threshold N , take $n \geq \max(2N + 1, N_0, N_1)$. Every index in W_n is at least N , so the window bound applies with $R = \left(\frac{3}{4}\right)^j M + 4\eta$, and

$$|e_n| \leq \frac{3}{4} \left(\left(\frac{3}{4}\right)^j M + 4\eta \right) + \eta = \left(\frac{3}{4}\right)^{j+1} M + 4\eta.$$

Choosing j large and η small makes the bound as small as we like, so $e_n \rightarrow 0$ and $c_n \rightarrow A$. □

The proof uses $\ln 2 < 1$ twice, in the boundedness step and in the contraction. That is where the window of half the indices matters: a window from $n/3$ to n would carry weight $\ln 3 > 1$, and the argument would prove nothing.

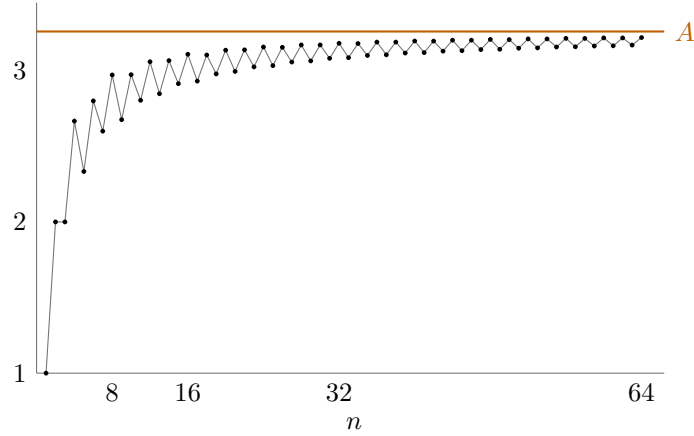


Figure 2. $c_n = na_n$ for $n \leq 64$, climbing towards $A = 1/(1 - \ln 2)$. In this range it rises at every even step, where $c_n = \frac{n}{n-1}c_{n-1}$, and falls at every odd step.

Remark 4. The rate is not proved here, but the numbers are regular. Iterating to $n = 2 \times 10^6$, $n(c_n - A)$ settles at -2.58870 for even n and -5.84759 for odd n . The two constants differ by A to the digits shown, which the even step explains: $c_{2m} - c_{2m-1} = c_{2m-1}/(2m-1) \approx A/(2m)$. In a continuous model, with $n = e^\tau$, identity (1) becomes the delay equation $c(\tau) = 1 + \int_{\tau-\ln 2}^\tau c$, whose perturbations $e^{\lambda\tau}$ satisfy $\lambda = 1 - 2^{-\lambda}$. The root $\lambda = -1$ is a perturbation of size $1/n$ that solves the homogeneous equation by itself, which suggests that the coefficient of $1/n$ depends on the whole early history of the sequence and has no closed form.

4. VERIFICATION

Lemmas 2 and 3, the boundedness of c_n , identity (2), the contraction argument and Theorem 1 are proved in Lean 4 with Mathlib [2], using only the standard axioms; Lemma 3 uses Mathlib's limit of $H_m - \ln m$. A Python program checks the window identity in exact rational arithmetic for $n \leq 3000$ and iterates to $n = 2\,000\,001$ for the remark, and catches three planted errors in the recurrence and the window. Code and Lean files are at [3].

ACKNOWLEDGMENTS

The question is by the Mathematics Stack Exchange user AAK, the constant was conjectured by Tian Vlasic, and the generating-function remark is Gary's. The proof, the computations and the Lean formalisation were carried out with Claude Opus 5.5 (Anthropic); an answer with this proof was posted to the question (answer 5150836). The author is responsible for the content.

REFERENCES

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